

Product Requirements Specification

Project NOVA Product requirements, Target Personas, and System Use Cases

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Revision History

Version	Date	Author	Summary of Changes
1.0	2026-06-28	Antigravity	Consolidate Product Requirements, Capability Catalog, Software Requirements, Success Criteria, and Use Case drafts.

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Purpose & Scope

This specification defines the functional boundaries, core target audience, capabilities, and system use cases for Project NOVA. It acts as the product requirements authority, validating that technical system architectures directly satisfy user needs.

Product Vision & Primary Goal

Vision

Build an AI Operating Platform that understands natural-language user intentions and safely orchestrates digital workflows.

Primary Goal

Substantially reduce manual keyboard and mouse interactions through secure, explainable, and intelligent desktop and web automation.

Core Principles

- **Natural Language First:** Input channels default to natural spoken or typed text queries.
- **User in Control:** Enforce confirmation barriers for high-risk actions.
- **Modular Capabilities:** Reuse capabilities as decoupled block elements.
- **Secure by Design:** Protect user credentials and system resources.
- **Explainable Actions:** Maintain transparent, auditable plans and event trails.

User Personas

NOVA addresses the following primary personas through targeted application skills:

8. **Developer:** Requires automation of command execution, workspace management, code compilation, and repository updates.
9. **Cybersecurity Student:** Requires safe orchestration of networks tools, script execution, and test sandbox environments.
10. **AI/ML Learner:** Requires integration of model runtimes, data pipelines, and package setup commands.
11. **Productivity Power User:** Requires automated file organization, calendar tracking, document summarization, and routine task triggers.
12. **General Computer User:** Requires simple voice-activated app launches and search routing.

Core Capabilities Catalog

NOVA's architecture is composed of these primary reusable capabilities:

- **Voice:** Captures spoken commands and generates natural speech feedback.
- **Vision:** Acquires displays, reads OCR layouts, and interprets visual UI elements.
- **Memory:** Stores, matches, and retrieves session context and preferences.
- **Planning:** Decomposes user goals into ordered tasks.
- **Context:** Synthesizes process state, window geometry, active text selection, and user history.
- **Desktop Control:** Launches processes, switches app focus, and emulates mouse clicks and keystrokes.
- **Browser Control:** Navigates pages, interacts with elements, manages cookies, and handles sessions.
- **Terminal Control:** Launches system shells and runs command scripts safely.
- **Workflow Engine:** Triggers, saves, and re-executes routine task plans.
- **Plugin System:** Integrates third-party APIs and custom tools.

Software Requirements

Functional Requirements

- **Voice Interface:** Support wake-word activation and natural transcription feedback loops.
- **Desktop & Browser Automation:** Programmatically orchestrate applications, window states, form elements, and file transfers.
- **OCR & UI Understanding:** Convert display layouts into structured bounding boxes and text blocks.
- **Reasoning & Planning:** Decompose natural inputs into structured execution plans.
- **Memory Retrieval:** Persist short-term conversation states and long-term workspace setups.

- **Security Permission Engine:** Validate actions against permissions before execution.

Non-Functional Requirements

- **Decoupled Modularity:** Every engine must interact via abstract interface classes.
- **High Testability:** Enforce coverage rules for all package modules.
- **Model Independence:** Abstract AI models (STT, TTS, LLMs) behind replaceable provider adapters.
- **Offline-First Support:** Route local capabilities to offline runtimes where practical to guarantee user privacy.

Initial Product Use Cases

- **UC-001: Voice Application Activation:** Launch system programs via voice queries.
- **UC-002: visual target Clicking:** Click buttons on screen by referencing text or UI shapes.
- **UC-003: Workspace State Restoration:** Restore application windows, sizes, coordinates, and file paths.
- **UC-004: Coding Workflow Execution:** Pull, compile, test, and commit a repository using developer profiles.
- **UC-005: Security Sandbox Scenarios:** Run security scripts and inspect sandboxed outputs.
- **UC-006: Web Research and Summarization:** Crawl target domains and generate summarized markdown results.
- **UC-007: Repetitive Tasks Automation:** Save and repeat a series of cursor clicks and text inputs.

Success Criteria

Version 1

- Decoupled, modular system architecture passes automated unit tests.
- High-accuracy voice transcription and audio generation.
- Desktop and browser emulators execute actions reliably under Windows.
- Basic local memory persists user workspace state JSON configurations.
- Custom tools register cleanly with the Tool Manager.

Long Term

- Fully autonomous task reasoning and visual grounding across complex OS environments.
- Self-learning memory logs that optimize task paths.
- Multi-device and cross-platform ecosystem support (Windows, macOS, Linux).